

The Mint: Multimodal Interoception Questionnaire

Final 33-item version · 11 facets · 3 metaclusters



Makowski, D., Neves, A., Benn, E., Bennett, M., & Poerio, G. (2026). *The Mint Scale: A Fresh Validation of the Multimodal Interoception Questionnaire and Comparison to the MAIA, BPQ and IAS* [Preprint]. <https://realitybending.github.io/InteroceptionScale/paper/manuscript.pdf>

Instructions given to respondents. “Please answer the following questions based on how accurately each statement describes you in general.”

Response scale. 7-point Likert scale, from 0 = *Disagree* to 6 = *Agree* (0 · 1 · 2 · 3 · 4 · 5 · 6). No items are reverse-scored.

Scoring. Facet scores are the **average** of their 3 items. Metacluster scores are the **average of all the items** belonging to that metacluster (i.e., 9 items for Interoceptive Awareness, 15 for Interoceptive Deficit, 9 for Visceroception).

Administration. Items are presented in a **random order**; the grouping below is for scoring purposes only and should not be shown to respondents. An attention-check item may be interleaved (“I can always accurately answer to the extreme left on this question to show that I am reading it”).

Hierarchical structure

Interoceptive Awareness	Expulsion Accuracy (ExAc) · Relaxation Awareness (RelA) · Sexual Arousal Sensitivity (SexS)
Interoceptive Deficit	Cardiorespiratory Confusion (CaCo) · Urointestinal Inaccuracy (UrIn) · Dermal Hypersensitivity (Derm) · Satiety Noticing (Sati) · Olfactory Contribution (Olfa)
Visceroception	Respiroception (Resp) · Cardioception (Card) · Gastroception (Gast)

Interoceptive Awareness 9 items · 3 facets

Expulsion Accuracy MINT_ExAc

- 1 I can always accurately feel when I am about to fart
- 2 I can always accurately feel when I am about to sneeze
- 3 I can always accurately feel when I am about to burp

Relaxation Awareness MINT_ReLA

- 4 I always feel in my body if I am relaxed
- 5 I always know when I am relaxed
- 6 My body is always in the same specific state when I am relaxed

Sexual Arousal Sensitivity MINT_SexS

- 7 During sex or masturbation, I often feel very strong sensations coming from my genital areas
- 8 My genital organs are very sensitive to pleasant stimulations
- 9 When I am sexually aroused, I often notice specific sensations in my genital area (e.g., tingling, warmth, wetness, stiffness, pulsations)

Interoceptive Deficit 15 items · 5 facets

Cardiorespiratory Confusion MINT_CaCo

- 10 Sometimes my breathing becomes erratic or shallow and I often don't know why
- 11 I often feel like I can't get enough oxygen by breathing normally
- 12 Sometimes my heart starts racing and I often don't know why

Urointestinal Inaccuracy MINT_Urin

- 13 I sometimes feel like I need to urinate or defecate but when I go to the bathroom I produce less than I expected
- 14 I often feel the need to urinate even when my bladder is not full
- 15 Sometimes I am not sure whether I need to go to the toilet or not (to urinate or defecate)

Dermal Hypersensitivity MINT_Derm

- 16 In general, my skin is very sensitive
- 17 My skin is susceptible to itchy fabrics and materials
- 18 I can notice even very subtle stimulations to my skin (e.g., very light touches)

Satiety Noticing MINT_Sati

- 19 I don't always feel the need to eat until I am really hungry
- 20 Sometimes I don't realise I was hungry until I ate something
- 21 I don't always feel the need to drink until I am really thirsty

Olfactory Contribution MINT_OLfa

- 22 I often check the smell of my armpits
- 23 I often check the smell of my own breath
- 24 I often check the smell of my farts

Visceroception 9 items · 3 facets

Respiroception MINT_Resp

- 25 In general, I am very sensitive to changes in my breathing
- 26 I can notice even very subtle changes in my breathing
- 27 I am always very aware of how I am breathing, even when I am calm

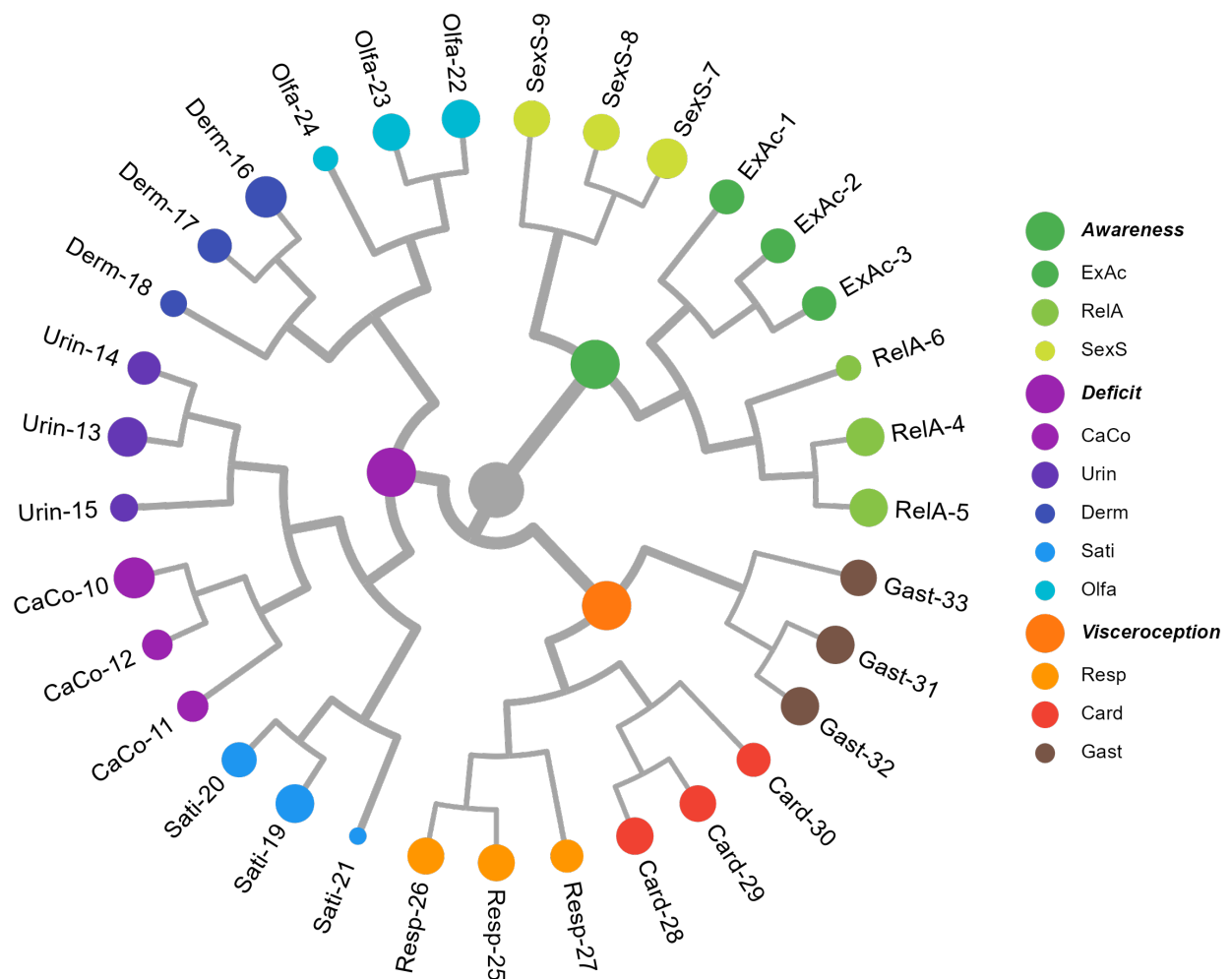
Cardioception MINT_Card

- 28 In general, I am very sensitive to changes in my heart rate
- 29 I often notice changes in my heart rate
- 30 I can notice even very subtle changes in the way my heart beats

Gastroception MINT_Gast

- 31 I can notice even very subtle changes in what my stomach is doing
- 32 In general, I am very sensitive to what my stomach is doing
- 33 I am always very aware of what my stomach is doing, even when I am calm

Structure of the 33 items



Hierarchical clustering of the final 33 items (Study 2, $N = 737$). Each leaf is an item, labelled by its facet and item number; node size reflects the item's cluster centrality (the Exploratory Graph Analysis equivalent of a factor loading). The three inner nodes are the metaclusters. This structure replicated the one obtained with hierarchical Exploratory Graph Analysis, supporting its consistency across methods.

Item variable names follow the pattern MINT_<Facet>_<item number> (e.g., MINT_ExAc_1, MINT_Gast_33), matching the experiment code and the analysis scripts. Internal consistency of the metaclusters in Study 2 ($N = 737$): Interoceptive Awareness $\omega = 0.85$, Interoceptive Deficit $\omega = 0.74$, Visceroception $\omega = 0.88$. Note that the *Olfa* facet showed lower structural stability within the *Deficit* metacluster and warrants further research.

Reference: Makowski et al., *Introducing the Mint: Validation of a Fresh Multimodal Interoception Questionnaire*. github.com/RealityBending/InteroceptionScale